Ch. 6 HW/CW

Principles of Microeconomics

September 30, 2025

1. At Fenway Park, home of the Boston Red Sox, seating is limited to about 38,000. Hence, the number of tickets issued is fixed at that figure. Seeing a golden opportunity to raise revenue, the City of Boston levies a per ticket tax of $5 to be paid by the ticket buyer. Boston sports fans, a famously civic-minded lot, dutifully send in the $5 per ticket. Draw a well-labeled graph showing the impact of the tax. On whom does the tax burden fall–the team's owners, the fans, or both? Why?

2. A market is described by the following supply and demand curves:

$$Q_S = 2P$$
$$Q_D = 300 - P.$$

   (a) Solve for the equilibrium price and quantity.

   (b) If the government imposes a price ceiling of $90, does a shortage or surplus (or neither) develop? What are the price, quantity supplied, quantity demanded, and size of the shortage or surplus?

   (c) If the government imposes a price floor of $90, does a shortage or surplus (or neither) develop? What are the price, quantity supplied, quantity demanded, and size of the shortage or surplus?

   (d) Instead of a price control, the government levies a $30 tax on producers. Does a shortage or surplus (or neither) develop? What are the price, quantity supplied, quantity demanded, and size of the shortage or surplus?

3. The market for an hour of tutoring is defined by the following supply and demand curves:

$$Q_D = 2,500 - 3P$$
$$Q_S = 5P - 100.$$

   If the government implements a price ceiling of $300, how much does the quantity of tutoring decrease?

4. The market for uber rides is defined by the following supply and demand curves:

$$Q_D = 150 - 5P$$
$$Q_S = 5P$$

   (a) If the government implements a price floor of $20, how much does the quantity of uber rides decrease?

   (b) If instead the government levied a $2 tax on uber drivers for ever ride sold, how much would the quantity of rides decrease?

   (c) If the government levied the same $2 tax on uber drivers for every ride sold, what would the incidence of the tax be? Please answer in percentages.

5. The market for a gallon of gas is defined by the following supply and demand curves:

$$Q_D = 600 - 150P$$
$$Q_S = 200P - 100$$

   The government implements a tax of $0.875 per gallon of gas sold on buyers.

   (a) What is the pre-tax equilibrium price and quantity in this market?

   (b) After the tax, what is the price consumers pay and the price producers receive?

   (c) What is the after-tax equilibrium quantity in this market?